

Camera calibration and re-calibration

Our general setup uses a surgical camera and fuses a preoperative 3D image, namely MRI or CT, with it to implement augmented reality surgical guidance. More information is found in our papers <https://encov.ip.uca.fr/publications/index.php> at EnCoV or on SURGAR's website <https://surgar-surgery.com/index.php/our-solution-2>.

Concretely, we need camera calibration to be available, which typically is matrix K and the distortion parameters. This calibration can be found using a calibration pattern such as a chessboard at the start of surgery. However, this is not convenient and is defeated by changing the camera configuration by focusing or zooming. Hence, we need to adapt the camera calibration dynamically during surgery.

Camera calibration and re-calibration, aka self- or auto-calibration, are well-understood in classical vision. We think that it would be interesting to implement and test these methods in the learning-based framework.

Deliverables:

1. A list of the important camera parameters in the perspective or projective camera model.
2. Specification of data type, constraints and ways to embed this model into a predictive scheme with neural networks.
3. Pseudo-code or Python code to define and test the two above points.